
Deep Active Learning Strategies Evaluation for Medical Image Classification on MedMNIST

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1. Specific Aims

Annotating 3D medical image data is expensive and time-consuming, which motivates the use of active learning (AL) to improve label efficiency. In this project, we study how different active learning strategies affect model performance on medical image classification tasks using the MedMNIST3D benchmark. In particular, we focus on how uncertainty estimation methods and batch selection strategies influence sample efficiency and training stability.

We begin by establishing a passive learning baseline, where a 3D ResNet-50 model is trained with randomly selected samples at each iteration. This provides a reference point for evaluating active learning methods and highlights the instability and inefficiency of random sampling in low-label regimes.

We then investigate uncertainty-based sampling using Monte Carlo dropout methods, including Bernoulli dropout and Gaussian dropout. By performing multiple stochastic forward passes, we estimate predictive uncertainty and use it to prioritize informative samples. We further analyze how different dropout mechanisms and noise levels affect both uncertainty quality and downstream classification performance.

To complement uncertainty-based approaches, we incorporate diversity-aware sampling through a design-of-experiments (DOE) based method (LapGOD), which selects samples to better cover the data manifold. We evaluate both standalone DOE initialization and hybrid strategies that combine DOE with uncertainty-based sampling, in order to assess whether diversity improves label efficiency.

Finally, we compare all methods under a consistent training protocol using metrics such as validation and test AUC across active learning rounds. We focus on differences in final performance, convergence behavior, and training stability, aiming to identify which strategies are most effective in practice under limited labeling budgets.

2. Background

Active learning (AL) is a framework designed to reduce labeling costs by iteratively selecting the most informative samples from an unlabeled pool for annotation[7]. Instead of relying on large amounts of randomly labeled data, AL prioritizes samples that are expected to provide the greatest improvement to the model[2]. This paradigm is particularly suitable for medical imaging tasks, where annotations are expensive and require expert knowledge.

A common approach in active learning is uncertainty-based sampling[15], where the model selects samples for which it is least confident. Typical strategies include least-confidence[6, 8], margin-based[10], and entropy-based selection[14], all of which rely on the predictive probability distribution of the model. These methods have been widely studied and are simple to implement, but their effectiveness depends heavily on the quality of uncertainty estimation, which is often unreliable in deep neural networks.

To address this limitation, Bayesian approximations such as Monte Carlo (MC) dropout have been proposed to improve uncertainty estimation in deep learning models[3]. MC dropout introduces stochasticity by performing multiple forward passes with dropout enabled at inference time, allowing the model to approximate a predictive distribution rather than a single deterministic output. The variance or entropy of these predictions can then be used as a measure of uncertainty. Different dropout variants, such as Bernoulli dropout and Gaussian dropout, provide alternative ways to inject stochasticity and may lead to different uncertainty characteristics.

In addition to uncertainty, diversity is an important consideration in batch active learning[13]. Selecting only the most uncertain samples can lead to redundancy, where many similar samples are chosen within a single batch. To mitigate this issue, diversity-aware sampling methods aim to ensure coverage of the data distribution. One class of such methods is based on design-of-experiments (DOE) principles[17], which select samples to optimize criteria such as representativeness or coverage of the feature space. In particular, Laplacian-based approaches (e.g., LapGOD)[12] incorporate the structure of the data manifold to guide sample selection.

3. Significance

These considerations are particularly relevant in clinical imaging applications, where data annotation is costly and often requires expert knowledge. For example, rib fractures can lead to severe complications, including respiratory failure and organ injury, and therefore require timely and accurate diagnosis [5]. In such settings, the availability of labeled data is often limited, especially for 3D medical images, making it challenging to train reliable deep learning models.

Active learning provides a practical approach to mitigate this limitation by selectively querying informative samples for annotation, thereby improving label efficiency. Instead of relying on large amounts of randomly labeled data, active learning focuses annotation efforts on samples that are expected to contribute more to model improvement. This is particularly useful in medical imaging scenarios, where labeling costs are high and data acquisition is expensive.

However, the effectiveness of active learning depends on the quality of the underlying sampling strategy, especially the ability to accurately estimate uncertainty and avoid redundant sample selection. Understanding how different uncertainty estimation methods influence active learning performance is therefore important for designing reliable annotation pipelines.

In this work, we provide an empirical evaluation of dropout-based uncertainty estimation methods within a controlled experimental setting. While our study is limited to a specific dataset and model architecture, the results offer insights into how different stochastic formulations affect sample selection and training stability in 3D medical image classification.

4. Experimental Design

We consider a pool-based active learning setting, where an initial labeled dataset $\mathcal{D}_L$ and a large unlabeled pool $\mathcal{D}_U$ are given. At each iteration, a batch of samples is selected from $\mathcal{D}_U$ according to a query strategy, labeled by an oracle, and added to $\mathcal{D}_L$. The model is retrained on the updated labeled set.

4.1. Uncertainty Estimation

We estimate predictive uncertainty using a Monte Carlo (MC) sampling framework with stochastic dropout. For each input sample, we perform multiple stochastic forward passes through the network and compute statistics over the resulting predictions.

4.1.1. Monte Carlo Estimation

Given an input sample x , we perform T forward passes with dropout enabled to obtain a set of predictions $\{\hat{y}^{(t)}\}_{t=1}^T$. The predictive mean is computed as:

$$\mu(x) = \frac{1}{T} \sum_{t=1}^T \hat{y}^{(t)}, \quad (4.1)$$

and the predictive variance is used as an uncertainty estimate:

$$\sigma^2(x) = \frac{1}{T} \sum_{t=1}^T \left(\hat{y}^{(t)} - \mu(x) \right)^2. \quad (4.2)$$

Samples with higher predictive variance are considered more uncertain and are prioritized for selection in the active learning process.

Predictive entropy was used to quantify uncertainty

$$H[y|x] = - \sum_c \mu_c(x) \log \mu_c(x). \quad (4.3)$$

4.1.2. Bernoulli MC Dropout

Bernoulli MC Dropout approximates Bayesian inference by enabling dropout at test time, where each unit is randomly dropped with a fixed probability. This introduces discrete stochasticity into the network and produces a distribution of predictions across forward passes.

4.1.3. Gaussian MC Dropout

Gaussian MC Dropout replaces binary dropout masks with multiplicative Gaussian noise. Specifically, each activation is scaled by $z \sim \mathcal{N}(1, \alpha)$, where α controls the noise level [9]. Compared to Bernoulli dropout, Gaussian dropout introduces continuous perturbations, which can result in smoother uncertainty estimates.

4.2. Initialization Strategy

Random selection This is the baseline for initial observed set selection. We select 200 samples randomly from the train set in the first round.

G-optimal DOE with Laplacian regularization This idea is based on LapGOD[1]. It is an DOE method which selects the data points that minimizes the maximum variance of the predicted values on the data manifold. In LapGOD, considering solving the linear function $y = \theta X$, the MLE estimation with Laplacian regularization term can be written as:

$$\theta = P^{-1}V y$$

$$\text{where } P = VV^T + \alpha XLX^T + \beta I.$$

Here, V refers to the observed set, X is the raw data and L is the Laplacian matrix. Here we use the binary Laplacian matrix generated by KNN with $k = 2$. And the objective function is:

$$F(V) = \min_V \max_{x_i \in X} \{x_i^T P^{-1} V V^T P^{-1} x_i\}.$$

Solving this optimization problem is computationally expensive since it includes many matrix inversion operations. So we try to solve it in a greedy way.

$$v_{k+1} = \arg \min_{v \in \mathcal{X} - V_k} \max_{x_i \in \mathcal{X}} \left\{ x_i^T (P_k + vv^T)^{-1} (V_k V_k^T + vv^T) (P_k + vv^T)^{-1} x_i \right\}$$

Using Sherman-Morrison-Woodbury formula, we can change the part including inversion operation to an iterative equation.

$$(P_k + vv^T)^{-1} = P_k^{-1} - \frac{P_k^{-1} vv^T P_k^{-1}}{1 + v^T P_k^{-1} v}.$$

4.3. Implementation Details

Data We evaluate our methods on the MedMNIST benchmark [18], a large-scale collection of lightweight medical image classification datasets derived from real-world medical imaging tasks. The dataset is designed to work with various classification tasks, including multi-class data, which we will use 3D FractureMNIST3D $28 \times 28 \times 28$ size image to classification task of distinguishing between different fracture conditions. The FractureMNIST3D is originated from a publicly available rib fracture CT dataset [4], containing CT scans of rib fractures collected from clinical imaging data. There are in total 1,370 samples in the dataset. And the dataset is split into training, validation and test set with a ratio of 7:1:2.

Model. We use a 3D ResNet-50 as the base classifier for all experiments. The model takes 3D volumetric images as input and outputs class probabilities over the three fracture categories.

Baseline. We use passive learning as the baseline, where both the initial labeled set and subsequent samples are selected randomly. This provides a reference for evaluating the effectiveness of active learning strategies.

Training. We initialize the labeled set with 200 randomly sampled instances to ensure stable early-stage training. At each active learning round, 20 samples are selected from the unlabeled pool and added to the training set. The model is trained for up to 5 epochs with a batch size of 32, and early stopping is applied based on validation loss.

Evaluation Metrics. We evaluate model performance using validation and test AUC across active learning rounds. Loss curves are also reported to assess training stability. For each method, we compare both final performance and the trajectory of performance as the number of labeled samples increases. This allows us to evaluate not only whether a method achieves higher final accuracy, but also whether it improves sample efficiency and reduces instability during training.

Code. Our code is public and available at <https://github.com/ziyuxli/automation>.

5. Results

5.1. Dropout Strategies

5.1.1. Bernoulli MC Dropout

Implementing Bernoulli Monte Carlo sampled dropout only improves the ResNet-50 model for dropout rates above 0.2 indicating increased dropout likely improves the trade off between regularization. Zeroing 10% has very little impact, and actually has lower accuracy in comparison to passive learning, only briefly doing better halfway through the active learning rounds as seen in top right plot of Figure 1. This is primarily due to the predictions across MC samples being too similar since 90 % of the nodes are retained in each layer and may have similar uncertainty. This is improved with stronger dropout values. Both the moderate (0.2) and high (0.5) dropout rates have better loss curves. Based on the difference in the loss curves in Figure 1, dropout rate of 0.5 has the least loss but is less accurate in comparison to zeroing 20% of the nodes. The accuracy curves for the moderate and high dropout rates remain relatively close for the validation AUC curve, but the 0.2 dropout rate generally has better results on the test set. It is possible that with a larger dataset, a 0.5 dropout rate could outperform the 0.2 dropout rate, but for this architecture a moderate dropout rate of 0.2 is the best trade-off between accuracy and loss. The overall better performance and curve of the moderate dropout rate indicates it is the best quantification of uncertainty estimates for the acquisition function.

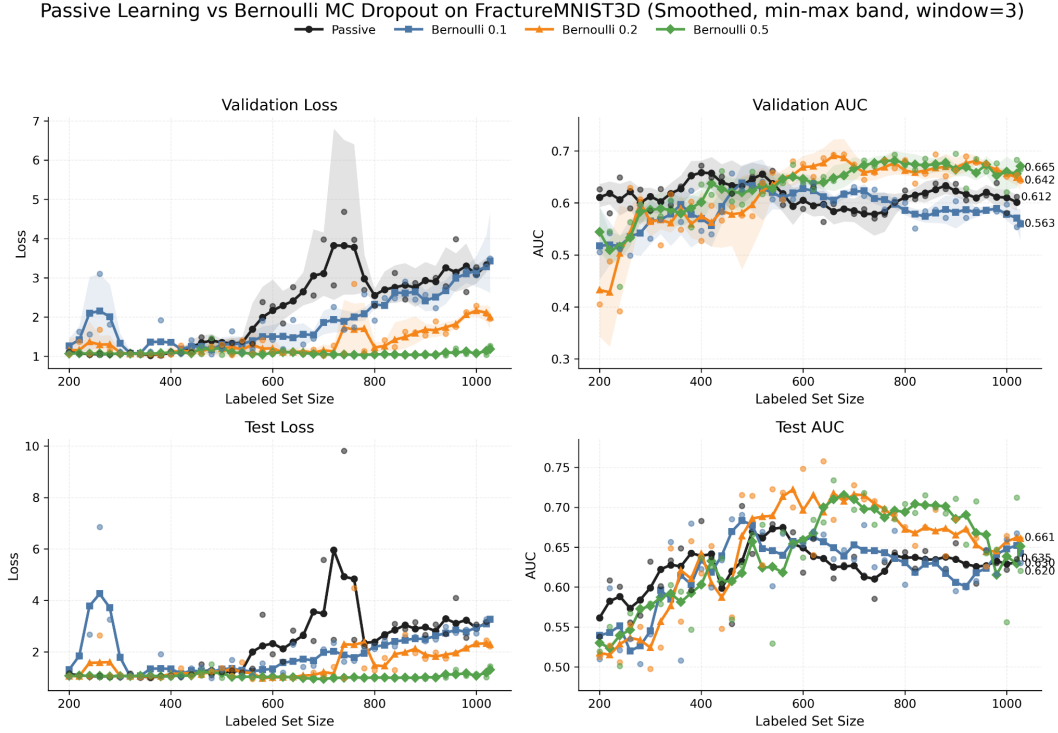


Figure 1: Smoothed validation and test performance for varying Bernoulli MC Dropout rates across active learning rounds. The shaded regions illustrate the range for each epoch. Black: passive learning. Blue(Low): Bernoulli dropout rate of 0.1, leaving 0.9 of the nodes. Orange(Moderate) : dropout rate of 0.2, leaving 0.8 of the nodes. Green(High): drops 0.5 of the nodes. Final AUC values are found on the far right for the validation (above) and test(below) AUC curves.

5.1.2. Gaussian MC Dropout

As shown in Figure 2, Gaussian MC Dropout consistently improves performance over passive learning across both validation and test metrics. The passive baseline exhibits substantial instability, with obvious spikes in both validation and test loss as the labeled set increases. In contrast, Gaussian dropout reduces these fluctuations, indicating more stable training dynamics.

Across different noise levels, all Gaussian dropout settings achieve higher AUC than passive learning. A moderate noise level (0.2) attains the best validation performance, reaching approximately 0.70 AUC, while a lower noise level (0.1) yields the highest AUC (approximately 0.73) in test set. This may suggest that moderate stochasticity improves uncertainty estimation during training, whereas lower noise may better preserve generalization on unseen data.

The highest noise level (0.5) shows slower convergence in early stages, but gradually improves as more labeled samples are added, eventually approaching competitive performance. This behavior indicates that excessive noise may hinder early optimization but still provides benefits in later stages by encouraging broader exploration of the data distribution.

Overall, these results demonstrate that Gaussian MC Dropout is an effective uncertainty estimation

method for active learning in 3D medical image classification, and that its performance is sensitive to the choice of noise level.

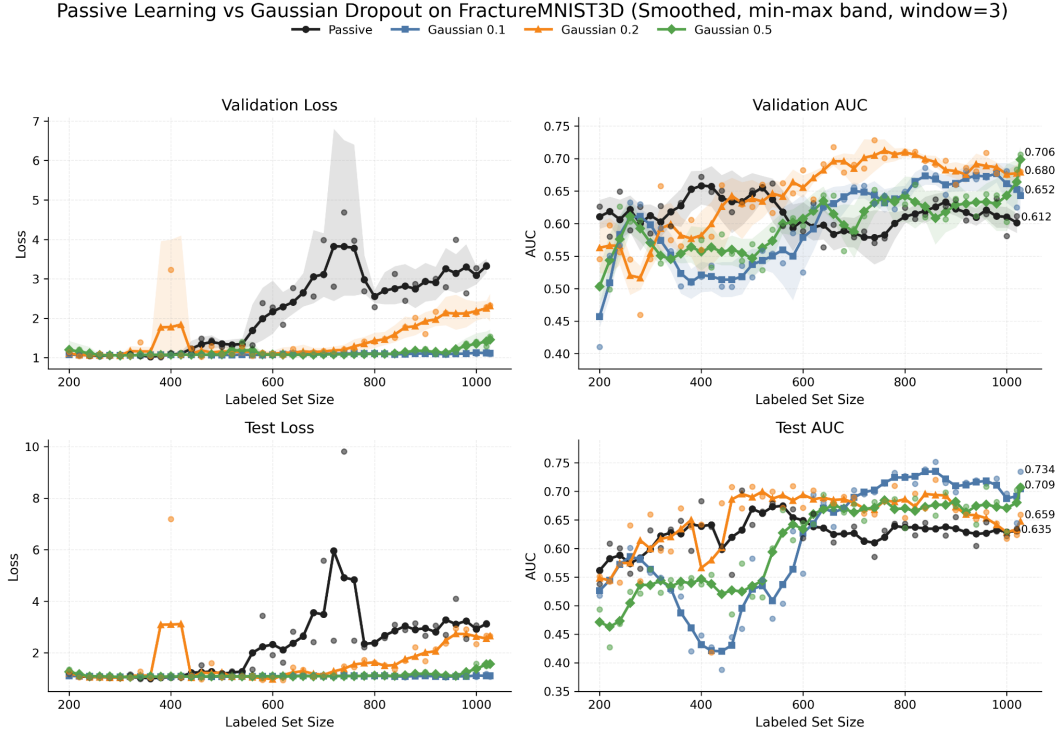


Figure 2: Smoothed validation and test performance for different Gaussian dropout noise levels compared to passive learning. The shaded regions indicate the range across epochs. Black: passive learning. Blue: Gaussian noise level 0.1. Orange: Gaussian noise level 0.2. Green: Gaussian noise level 0.5. Final AUC values are found on the far right for the validation (above) and test (below) AUC curves.

5.1.3. Bernoulli vs. Gaussian MC Dropout

In general, Gaussian MC dropout yielded lower loss on both the validation and test data and slightly outperforms Bernoulli MC dropout overall. This is more apparent when comparing the Test AUC curves as Bernoulli peaks in performance and then approaches values similar to those of passive learning. The Gaussian dropout performing better indicates that it is more useful to the acquisition function, thus leading to better sample selection. This result is within reason, as Gaussian Dropout scales activations via multiplicative Gaussian noise, whereas Bernoulli dropout zeros out entire nodes. In other words, Bernoulli dropout ends up adding more noise as the labeled set increases. As noise is already a challenge with this dataset, the Gaussian dropout produces a smoother approximation and uncertainty estimates. Since Bernoulli dropout randomly removes a certain number of nodes in the network layers, this makes it more difficult to handle borderline or noisy fractures in comparison to Gaussian dropout.

5.2. Initialization Strategy

5.2.1. LapGOD initialized passive learning

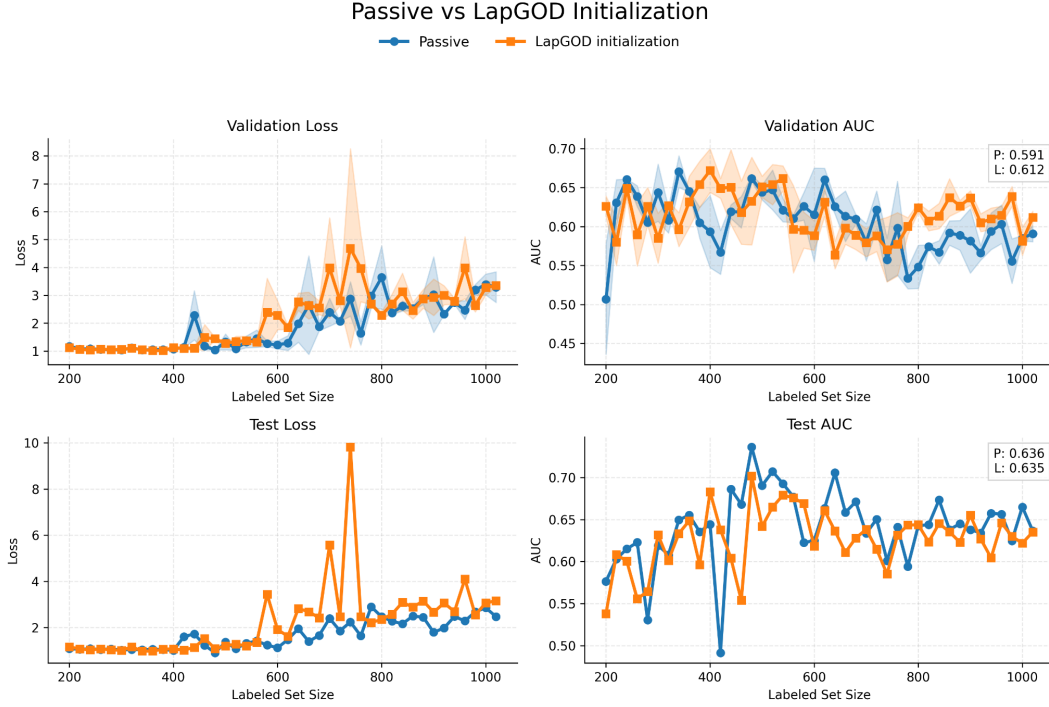


Figure 3: LapGOD initialization vs Passive Learning. The shaded regions indicate the range across epochs. Blue: Plain passive learning, observed set initialized randomly. Orange: Passive learning with LapGOD initialized observed set. Final AUC values are found on the far right for the validation (above) and test (below) AUC curves

As shown in Figure 3, the two methods show no substantial difference in performance. The AUC scores are nearly identical (Validation: 0.591 vs. 0.612; Test: 0.636 vs. 0.635), and both exhibit similar loss trends across labeled set sizes. This suggests that the initialization introduced by LapGOD does not provide a significant advantage over passive learning, likely because passive learning is less dependent on optimization path and therefore LapGOD initialization does not bring significant improvement.

5.2.2. LapGOD initialized Bernoulli MC Dropout

In Figure 4, LapGOD initialization shows a clear advantage over randomly initialized Bernoulli dropout. LapGOD achieves notably higher AUC scores on both validation (LapGOD(L): 0.612 vs. Random(P): 0.568) and test sets (LapGOD(L): 0.635 vs. Random(P): 0.578). And it also maintains stable loss, particularly in the early training stages. This suggests that when combined with an active learning strategy like Bernoulli dropout, LapGOD can meaningfully guide the selection of more informative labeled samples, leading to better model performance.

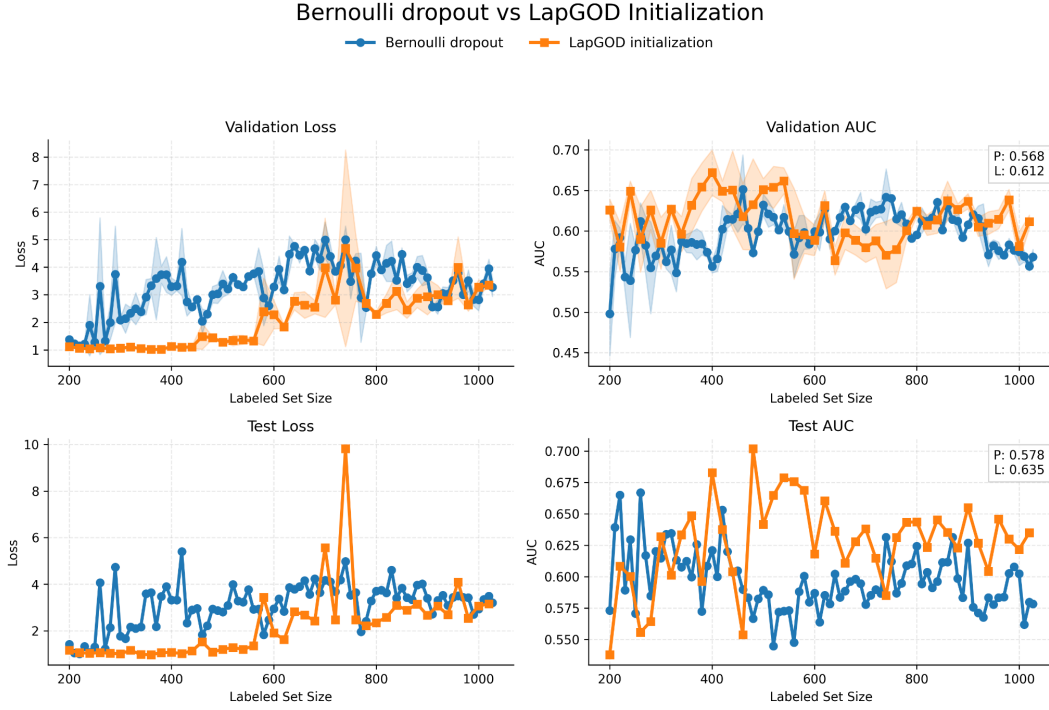


Figure 4: LapGOD initialized Bernoulli dropout vs Randomly initialized Bernoulli dropout. The shaded regions illustrate the range for each epoch. Blue: Plain Bernoulli dropout, observed set initialized randomly. Orange: Bernoulli dropout with LapGOD initialized observed set. Final AUC values are found on the far right for the validation (above) and test (below) AUC curves.

6. Discussion

6.1. Conclusions

The goal of this project was to explore improving the diagnostics of rib fractures by evaluating a potential DOE-based active learning approach. To establish the better dropout strategy to quantify uncertainty, Bernoulli Monte Carlo dropout and Gaussian Monte Carlo Dropout were compared on the base model and with the LapGOD DOE initialization. The goal was to see if implementing LapGOD DOE could overcome the difference in accuracy between the dropout method that yielded worse results overall. Moderate zeroing of nodes performed the best for Bernoulli MC dropout, and Gaussian MC dropout performed the best with the least amount of gaussian multivariate noise. As anticipated, both dropout methods improved the performance of the model in comparison to passive (random) approach, and the LapGOD initialization strategy further boosted the performance. Since LapGOD is originally designed to be compatible with standard MC dropout, we hypothesized that a LapGOD DOE + Bernoulli MC dropout implementation could potentially out-perform the Gaussian MC Dropout. However, this was not the case and while LapGOD improved the performance of the base model in comparison to the passive learning approach, it did not help the Bernoulli ResNet-50 model to overcome the performance of the Gaussian MC dropout model. These results were interesting because, but within reason, and Gaussian MC dropout is better suited to this particular architecture.

6.2. Future Directions

One major change we would make is determining how to implement the appropriate math to ensure compatibility with Gaussian MC dropout and LapGOD DOE initialization. One of the main challenges in the project was attempting to try and rework the LapGOD design to work with both Bernoulli MC dropout, as is the standard, and Gaussian dropout in a modular manner. In the future the optimal model found can be further expanded upon by using a transfer learning based approach. By implementing an ensemble deep learning model that integrates the ResNet-50 model and EfficientNetB3 models, has been shown to yield a more discriminative feature space [11]. An increase in robustness may result in better classification accuracy as seen in prior literature. In addition, a better acquisition function can be implemented, such as CLINICAL, which is a method designed to handle imbalanced medical image data and could be used to handle rare or more difficult images of rib fractures [16]. The combination of these two approaches can improve the robustness, accuracy, and reduce the amount of data needed to produce classifications of fracture image data. This can then be used as a foundation to set prescription and/or treatment guidelines to alleviate pressure on medical staff, especially during high patient volumes.

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